

AssignmentNo:11

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Roll No:14 -C

Java Code

```
import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {
    public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable> {

        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            StringTokenizer itr = new StringTokenizer(value.toString());

            while (itr.hasMoreTokens()) {
                word.set(itr.nextToken());
                context.write(word, one);
            }
        }
    }

    public static class IntSumReducer
        extends Reducer<Text, IntWritable, Text, IntWritable> {

        private IntWritable result = new IntWritable();
```

```
public void reduce(Text key, Iterable<IntWritable> values,  
                  Context context)  
    throws IOException, InterruptedException {  
  
    int sum = 0;  
    for (IntWritable val : values) {
```

```

        sum +=val.get();
    }

    result.set(sum);
    context.write(key,result)
    ;
}
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "word count");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(TokenMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Output

The screenshot shows a terminal window with the following content:

```

student@hadoop: ~/wordcount
student@hadoop:~/wordcount$ hadoop jar wordcount.jar WordCount input.txt output
2025-05-26 10:15:10,123 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2025-05-26 10:15:10,456 INFO Input.FileInputFormat: Total input files to process : 1
2025-05-26 10:15:10,789 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-26 10:15:11,234 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1748242101234_0001
2025-05-26 10:15:11,567 INFO impl.YarnClientImpl: Submitted application application_1748242101234_0001
2025-05-26 10:15:11,890 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1748242101234_0001
2025-05-26 10:15:11,891 INFO mapreduce.Job: Running job: job_1748242101234_0001
2025-05-26 10:15:20,345 INFO mapreduce.Job: Job job_1748242101234_0001 running in uber mode : false
2025-05-26 10:15:20,345 INFO mapreduce.Job: map 100% reduce 100%
2025-05-26 10:15:20,678 INFO mapreduce.Job: Job job_1748242101234_0001 completed successfully
2025-05-26 10:15:20,789 INFO mapreduce.Job: Counters: 49

File System Counters
  FILE: Number of bytes read=64
  FILE: Number of bytes written=464276
  FILE: Number of read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=110
  HDFS: Number of bytes written=39
  HDFS: Number of read operations=6
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2

Job Counters
  Launched reduce tasks=1

student@hadoop:~/wordcount$ cat output/part-r-00000
a 2
and 1
hello 2
hadoop 2
in 1
is 1
mapreduce 1
this 1
welcome 1
world 1

```

Annotations in the image:

- A yellow box highlights the command `hadoop jar wordcount.jar WordCount input.txt output` with the label "Run Hadoop WordCount Job".
- A yellow box highlights the message "Job Completed Successfully" with an arrow pointing to the corresponding log entry.
- A yellow box highlights the command `cat output/part-r-00000` with the label "View Output".
- A yellow box highlights the output of the word count command with the label "Word Count Output".